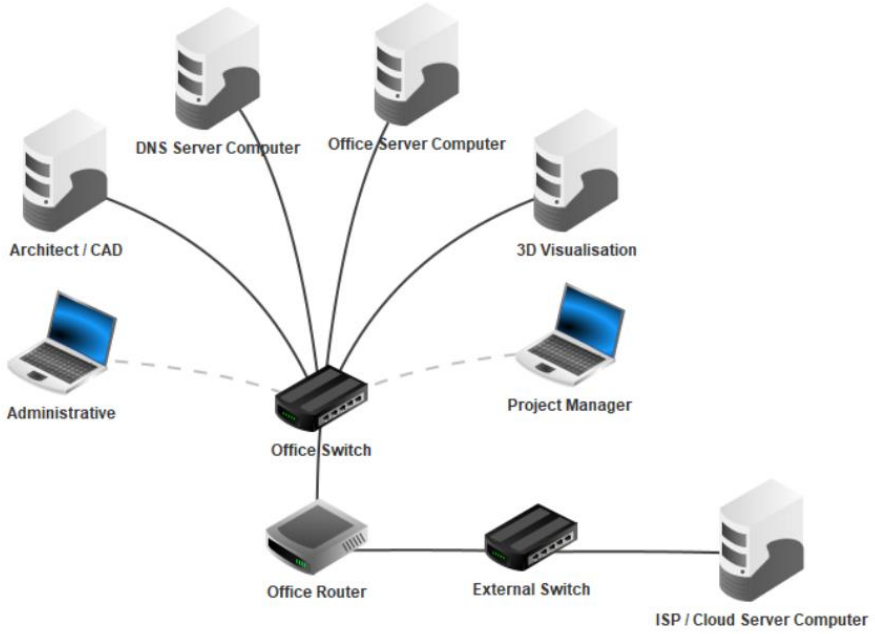
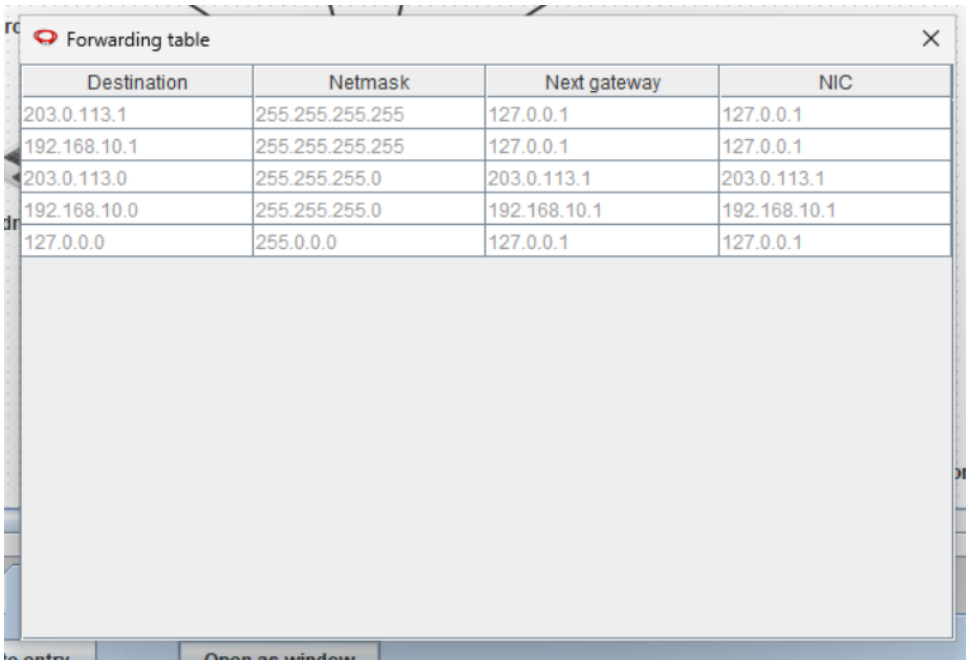
Project 3: NorthWest Architecture Studio Network

Topology:

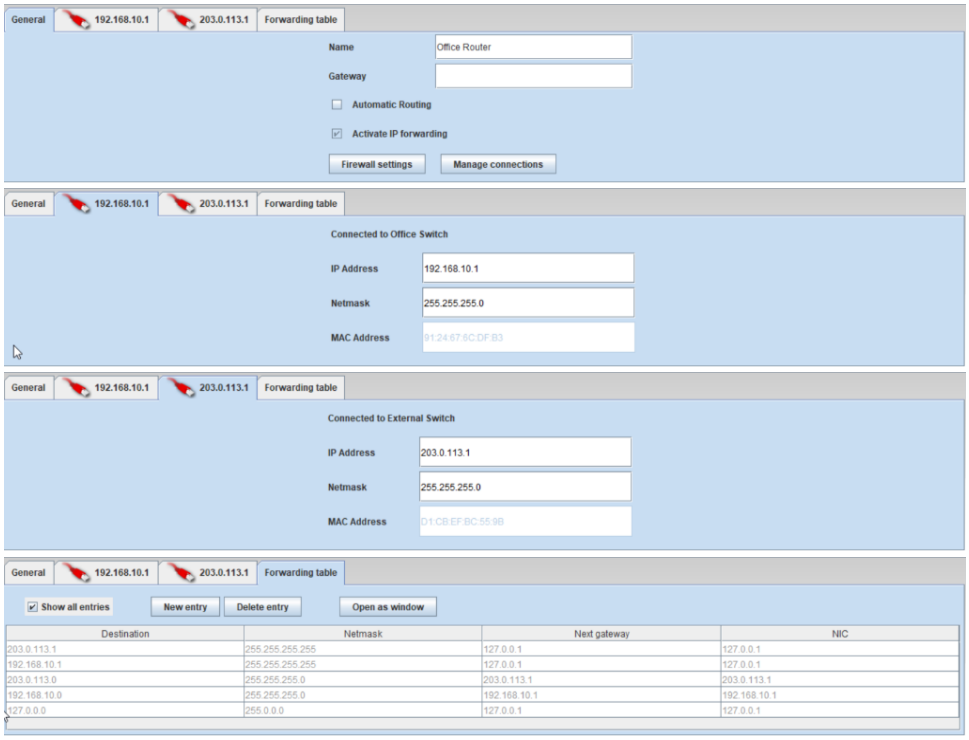


IP Addressing Table:

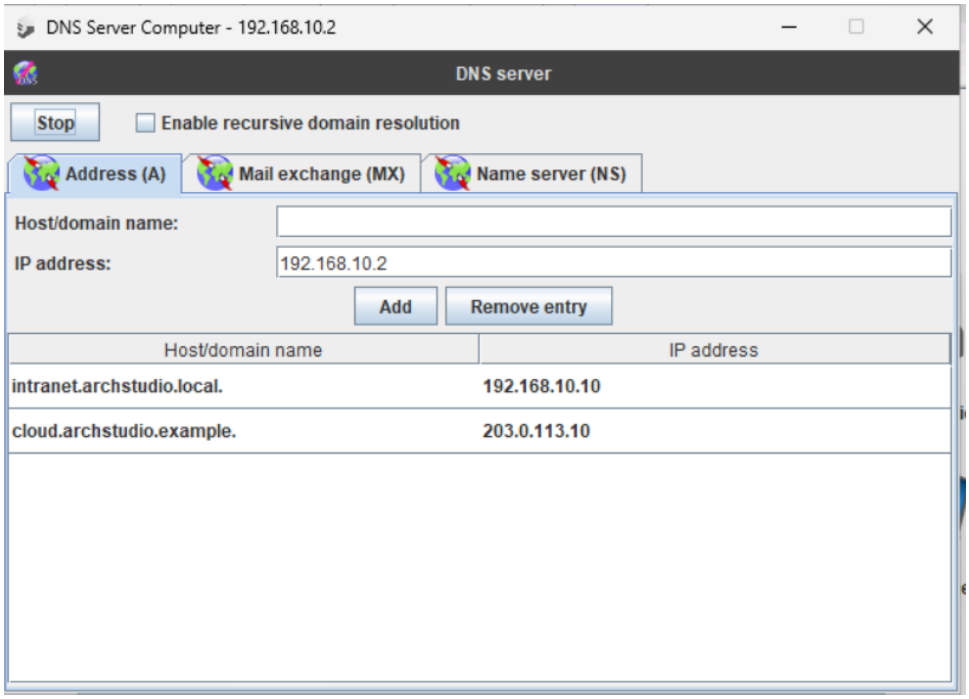

Forwarding table

Destination	Netmask	Next gateway	NIC
203.0.113.1	255.255.255.255	127.0.0.1	127.0.0.1
192.168.10.1	255.255.255.255	127.0.0.1	127.0.0.1
203.0.113.0	255.255.255.0	203.0.113.1	203.0.113.1
192.168.10.0	255.255.255.0	192.168.10.1	192.168.10.1
127.0.0.0	255.0.0.0	127.0.0.1	127.0.0.1

Router-Interface Screenshot:

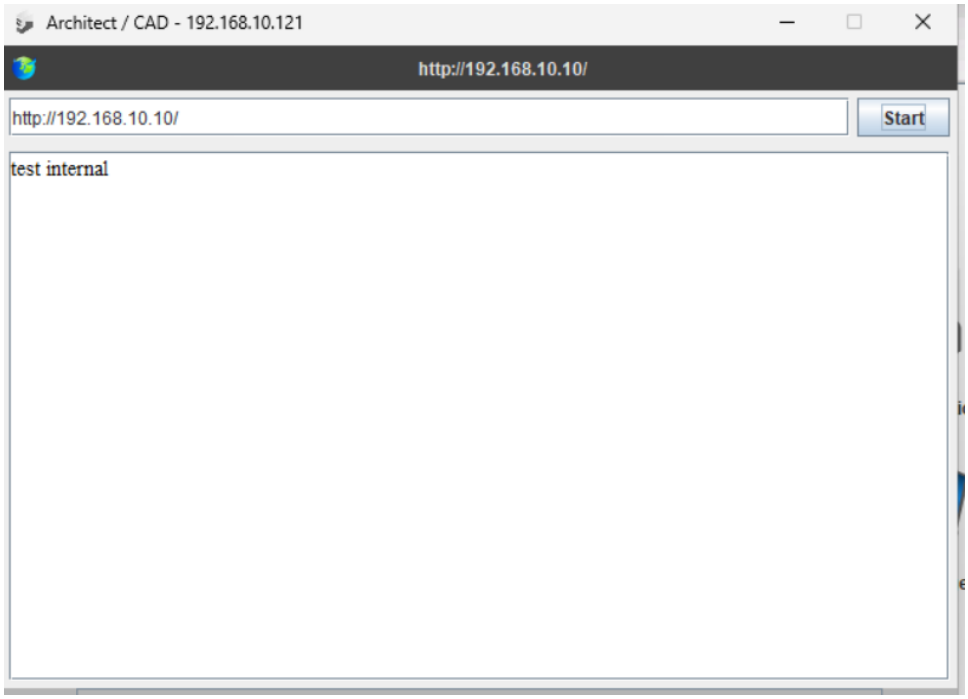


DNS-records screenshot:

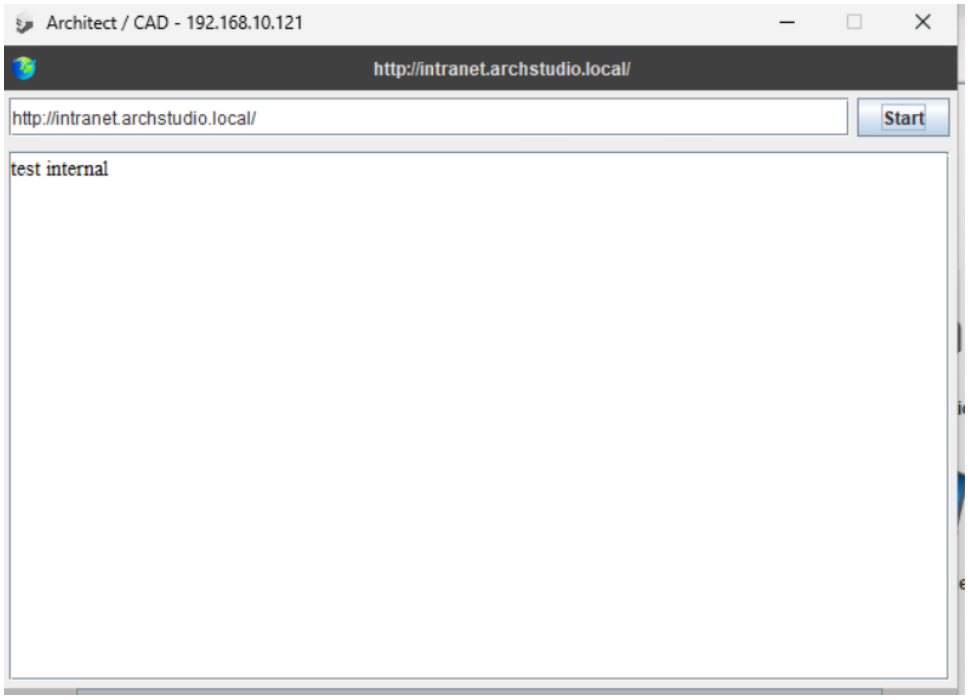


Internal Web Pages (from Architect / CAD Computer):

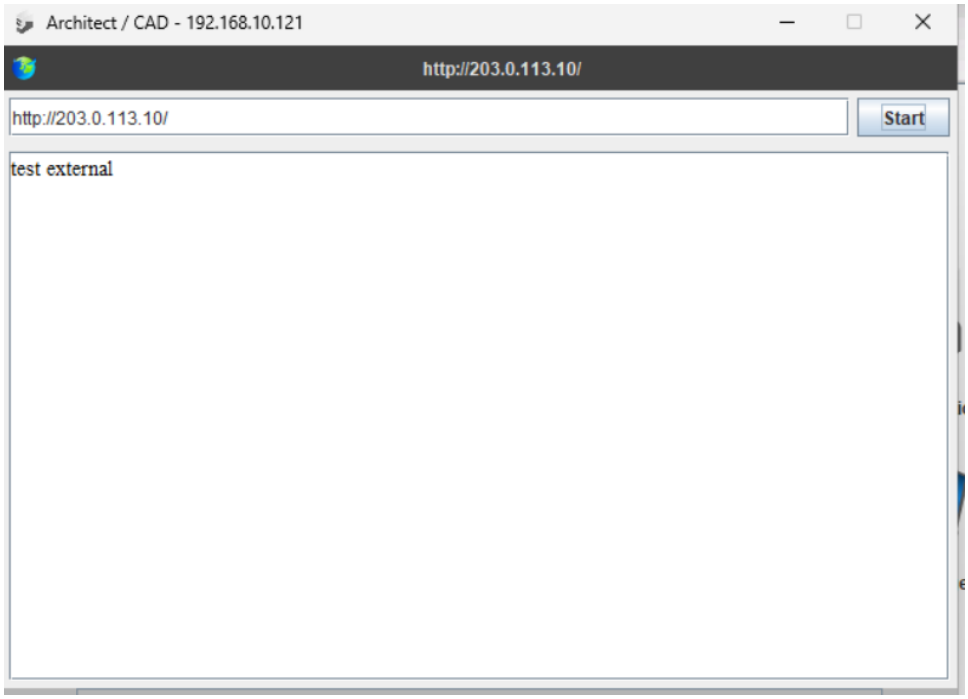
- Internal Web by IP



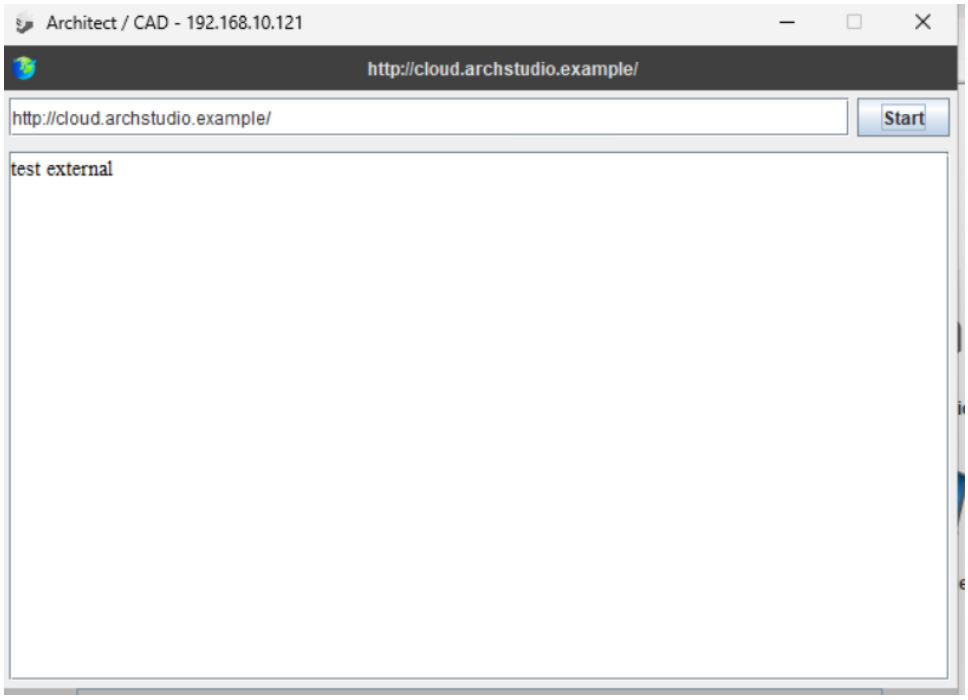
- Internal Web by name



- External Web by IP

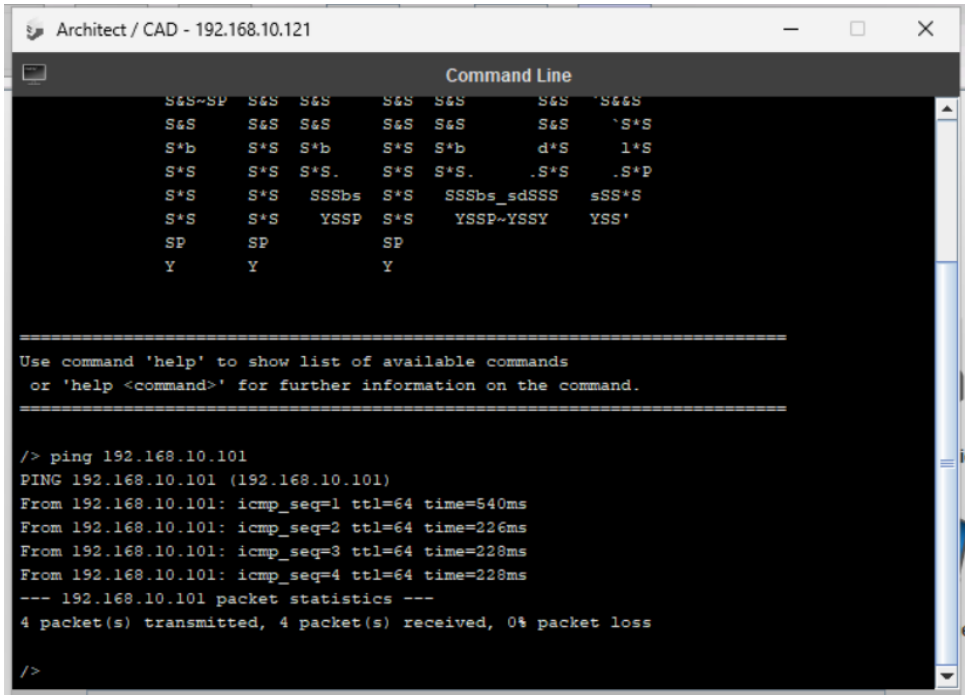


- External Web by name

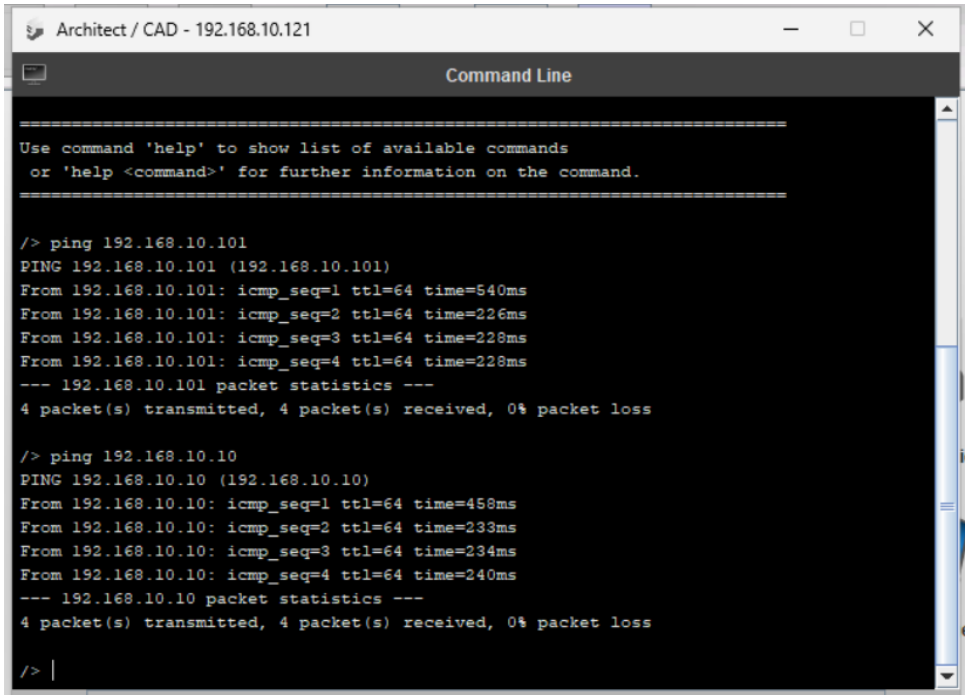


Ping Tests (from Architect / CAD Computer):

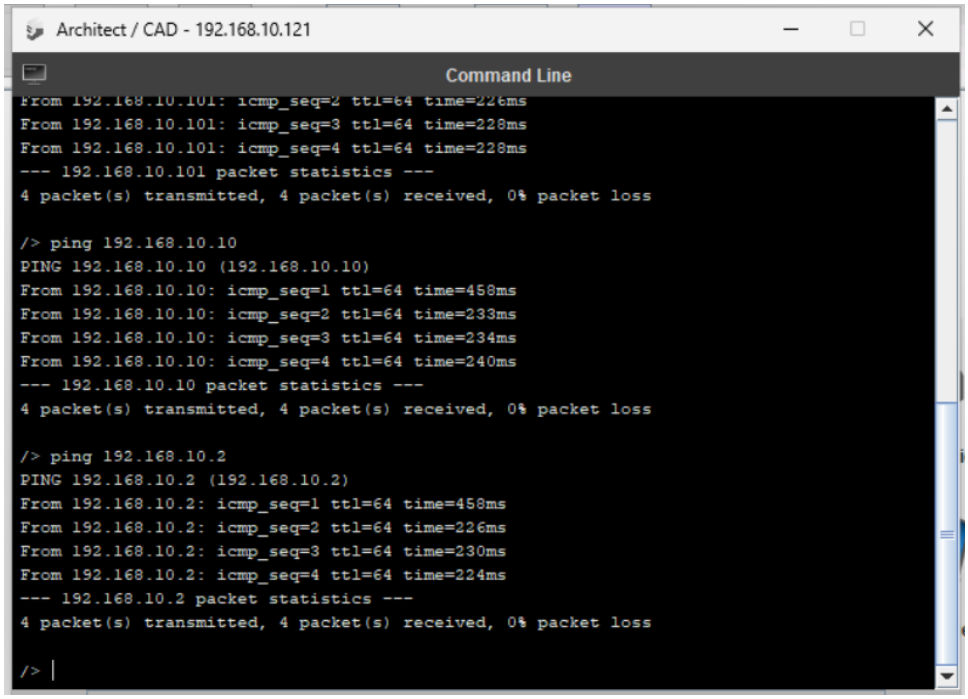
- LAN PC-to-PC



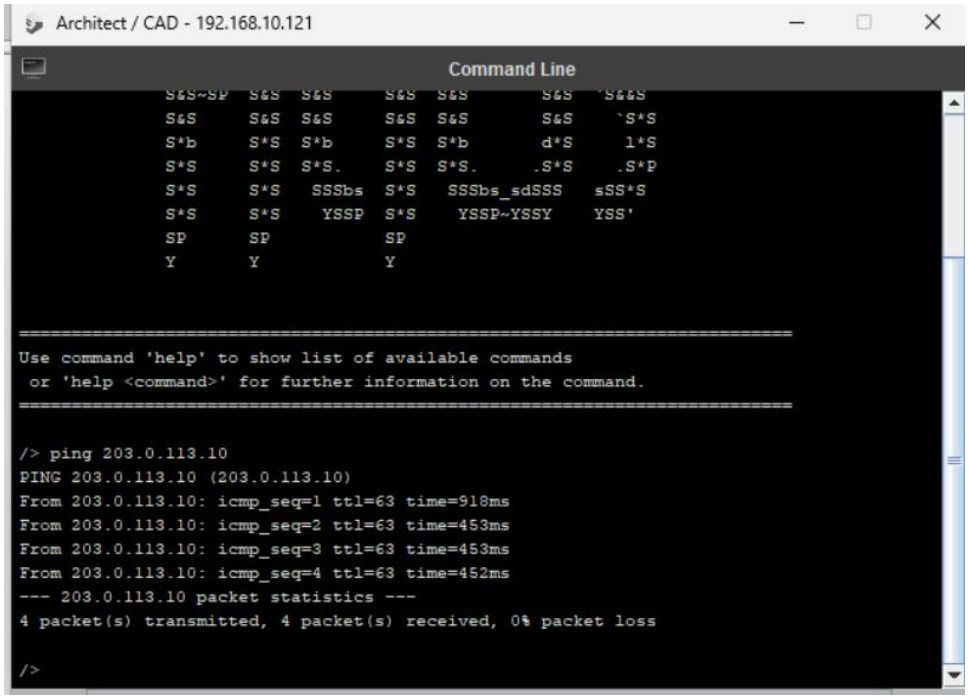
- PC to Office Server



- PC to DNS Server



- External Server Ping



Protocol Table with explanations

Concept	Where it appears	Explanation
TCP/IP	All device communication	Transmission Control Protocol/Internet Protocol: The basic set of rules that allows computers and devices to communicate across the internet and networks. IP handles the addressing/routing by breaking messages into smaller pieces (packets), and gives each a destination address. TCP basically puts these pieces back in order, and ensures data arrives accurately without errors/packet loss.
IP addressing	Every device has a unique IP	IP addressing is used to assign a unique number to every device connected to a network. This is so this device can send and receive data without sending it to the wrong device, and receiving data not meant for them.
Subnet mask	255.255.255.0 on both networks	A Subnet mask splits the IP address, showing the host part and the network part. e.g. 192.168.10.1 Network part – 192.168.10 Host part - .1
Gateway	Client PCs use 192.168.10.1	A gateway acts as a bridge between networks, allowing data to be transferred through this bridge. For example, between your home network and the internet.

		It guides data to where it should go, whilst also adhering to security features like firewalls, proxy, encryption, etc.
DNS	intranet.archstudio.local etc.	Domain Name System essentially converts an easily recognizable website name like youtube.com into an IP address. This is because the computer doesn't understand what 'youtube.com' is, so DNS turns it into an IP address that the computer can read, which then allows the user to be taken to the website.
HTTP	Opening pages in the browser	HTTP is used to transfer webpages, images, and data across the internet. The process includes your browser sending a request to a website, and the website sending a response back that either shows you the page or an error.
ICMP / ping	Ping tests	A ping basically tests if a device is reachable, and can send/receive data. It displays information like latency, packet loss, and TTL.
Client-server	CAD PC → Office/Cloud Server	A client-server is where the client (your device) asks for data/services from a server over a network. The process is essentially requesting for something, server processing this request and choosing whether or not to give it, then responding by giving the service or not.

LAN vs External Network

A LAN (Local Area Network) is a private network covering a single area, for example a home, office, etc - it connects devices like computers, phones and printers. However, an external network, also referred to as a WAN (Wide Area Network), connects multiple LANs together to allow devices to access data/servers/websites over very large distances, even worldwide.

A client-server is where the client (your device) asks for data/services from a server over a network. The process is essentially requesting for something, server processing this request and choosing whether or not to give it, then responding by giving the service or not.